Resilience to Climate Change: Challenges for the Baltic States

9

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Abstract

Climate resilience is the ability to adapt to changing climatic conditions, mitigate the effects of climate change, and effectively address the associated risks. The effects of climate change pose a major threat to the ecosystems, societies, and economies of small states. In order to build climate resilience, it is necessary to assess and identify the risks posed to countries by climate change. In this context, the most vulnerable sectors identified in the Baltic States are: natural ecosystems, food, health, infrastructure, and the economy. These sectors are vulnerable to increasing extreme weather events, rising Baltic Sea water levels, changes in temperature and precipitation as well as other climate changes. Therefore, this chapter reveals the resilience to climate change in the Baltic States through an analysis of the effects of climate change and identification of key changes and possible solutions through the concepts of absorptive, adaptive, and transformative capacities. The analysis shows that the Baltic States are focusing on building adaptive resilience capacity through long-term adaptation strategies such as restoring natural ecosystems, developing green urban areas, investing in research, strengthening governance structures responsible for managing climate crises, developing a financial support system, etc. These and other identified strategies and measures provide the basis for reducing the dependence of the Baltic economy on vulnerable sectors, maintaining the stability of natural ecosystems, public health, and quality of life.

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9.1 Introduction

Climate change is one of the most important challenges facing modern society. It is a continuous process that takes place in an ecosystem and has significant consequences for all elements of the system (natural, social, economic, environment, human quality of life, etc.). As human activities have led to global warming, which is already an irreversible process (EEA, 2024), the concept of climate resilience is increasingly being explored, which essentially encompasses two distinct strategies: adaptation to the effects of climate change (i.e., absorptive and adaptive capacities) and mitigation through the identification of measures to reduce climate pollution (transformative capacity) (Pivić et al., 2022; Issaka et al., 2023; Motevalli et al., 2023). Climate resilience is not only about adapting a country's infrastructure and economy, but also about the ability of society as a whole to react and make appropriate decisions in the face of changing conditions. The impacts of climate change are systemic and encompass social, economic, infrastructural, and political factors, i.e. they pose a threat to food supplies, the energy sector, biodiversity, coastal settlements, and human health.

For the Baltic States (Lithuania, Latvia, and Estonia), these threats are also relevant because of their dependence on natural ecosystems and their resources, changing conditions for agricultural development, rising Baltic Sea water levels, changes in biodiversity, etc. In the region, this is most pronounced in economic terms, as it is directly linked to climatic conditions, which have recently been changing, posing challenges such as erratic rainfall, prolonged droughts, the emergence of new pests and invasive species, and changing climate zones. Managing these threats therefore requires the implementation of a range of adaptation measures to increase resilience to climate change.

This chapter analyses the concept of climate resilience, identifies the main risks and effects of climate change, details the impacts of climate change on the Baltic States, and discusses possible strategic solutions to enhance the resilience capacities (absorptive, adaptive, and transformative) of these countries.

9.2 The Concept of Climate Resilience

More than two hundred years ago, the main causes of climate change were natural (solar cycles, changes in atmospheric composition, movement of tectonic plates, volcanic eruptions, etc.), but in the last two centuries, the root causes of climate change have been linked to human activity. This is associated with the onset of industrialisation: various gases released into the atmosphere have exacerbated the greenhouse effect, whereby gases in the Earth's atmosphere transmit the Sun's rays but trap heat, which results in temperatures in the lower atmosphere that are much higher than they would be if there were no clouds and greenhouse gases in the Earth's atmosphere. Of all greenhouse gases, carbon dioxide (CO2) is the largest contributor to the warming of the Earth's climate, with a significant increase in atmospheric concentrations compared to pre-industrial times (Škiudas et al., 2020).

Increases in these types of emissions are linked to to industrial and urban growth, deforestation, intensive agricultural development, rising transport activity, and changing lifestyles and consumption patterns across and within different regions and countries. Every year, human activities release around 40 gigatons (billion tonnes) of carbon dioxide emissions into the atmosphere (NOAA Climate.gov, 2024).

Europe is currently the fastest warming continent, with the continent warming at about twice the rate of the rest of the world since 1980. The year of 2023 was the hottest year globally in more than 100,000 years and 1.48 °C above pre-industrial levels. Although the change presented seems relatively small, scientists say that 2 °C of global warming will trigger irreversible loss to Earth's ice sheets, mountain glaciers and snow, sea ice, permafrost, and polar oceans (ICCI, 2023). This could lead to climate change spiralling out of control and irreversible changes to the planet. In this context, countries and societies are looking for solutions to increase resilience to climate change, to mitigate potential impacts and to adapt to the situation that already exists.

When analysing resilience in the context of climate change, it is important to recognise that it is a complex system with many elements. Aldunce et al. (2014) argue that it is primarily a system of social and ecological components, with virtually no boundaries, and that the systems are not isolated, but rather interact with each other and with other systems. Climate resilience is therefore a multidimensional concept that is defined as a process, ability, or outcome of withstanding the impacts of climate change and successful adaptation to the environment, despite dangerous or adverse conditions (Motevalli et al., 2023). Researchers (Issaka et al., 2023; Jayakrishnan et al., 2023) argue that the basic definition of climate resilience includes the following three key attributes: absorptive capacity, adaptive capacity, and transformative capacity. Absorptive capacity refers to the ability to withstand shocks and maintain functionality, while adaptive capacity refers to the ability to adapt and reorganise in response to changing conditions. The two are very closely linked, as according to Cheong and Assenova (2021), adaptive capacity depends on the ability of a community or an institution to assimilate new information and to recognise and relate new knowledge to existing experience. Transformative capacity refers to the potential for fundamental changes that increase long-term resilience. In summary, resilience is defined as the ability of a system to respond to temporary adverse events (shocks), such as climate variability, or long-term adverse trends (stresses), such as climate change, by being able to absorb, adapt and even transform (Isaaka et al., 2023). These areas are closely interlinked and can be realised in the long term through monitoring, analysis, and concrete action plans. This requires assessing climate change through a spatial-systemic and temporal lens (Resilience Alliance, 2010). The spatial-systemic dimension is important because changes in larger systems lead to changes in smaller ones.

But even in small systems that are part of a larger system, change can come from within. For example, when small parts of a coral reef are degraded, they can jeopardise the resilience of a much larger area by disrupting the access of coral larvae to the surrounding parts of the reef, which can lead to the degradation of the larger reef

system. Assessing the impacts of climate change or its mitigation cannot be limited to a specific area or a single factor, as any ecosystem element is part of a larger system. Therefore, when forecasting, mitigating, or managing risks, it is appropriate to analyse not only climatic factors, but also economic, social, or geopolitical factors with respect to the specificity of the situation under analysis. The time indicator is the result of an analysis of past actions (e.g. analysing natural disasters) and future actions (e.g. predicting climate change). Analysing changes in the system over time can help assess past conditions and modules more accurately, and develop future scenarios.

9.3 The Impact of Climate Change on the Environment and Society

Climate change impacts society and the environment. Durán-Sandoval et al. (2023) distinguish between the following two types of consequences of climate change: the primary consequences are physical (e.g. precipitation, increased frequency of extreme weather events, changes in temperature and sea level), whereas the secondary consequences are more varied and more difficult to predict, such as ecological, social, and economic consequences. The severity of effects also varies geographically, depending on the vulnerability of society and its capacity to adapt or mitigate the consequences through various social, economic, or political means and approaches.

The impact and irreversibility of the effects of climate change are well established and will be felt for decades. It is argued that the consequences can no longer be avoided even if no greenhouse gases are emitted. It is therefore essential to assess the risks and impacts associated with climate change. Five key areas of vulnerability to climate change, namely, natural ecosystems, food, health, infrastructure, and the economy (finance) (Table 9.1) have now been identified and should be the focus of national policy makers and decision-makers.

Natural ecosystems are already experiencing widespread and rapid changes: temperatures are rising, global mean sea level has risen by about 0.20 m, extreme events such as heat waves, tropical cyclones and droughts are becoming more frequent, hurricane winds and heavy rainfall are forming, increasing evaporation is threatening droughts and water scarcity, yields are declining, pests are increasing, and so on. The systemic effects of these factors are leading to societies facing severe food shortages and water scarcity, declining agricultural productivity; increased mortality from floods, droughts, and storms; the spread of new infectious diseases and changes in biodiversity; high rainfall leading to massive landslides that threaten human settlements; and the flooding of residential areas, destruction of infrastructure, etc., due to rising seawater and storm surges (Guzman et al., 2009).

Food. Climate change, temperature and precipitation shifts across climate zones are changing the conditions under which crops are grown. It is estimated that the gross factor productivity of global agriculture has declined by 21% since 1961, and

Table 9.1 Risks in Europe from climate change effects by area of impact (based on European climate risk assessment: executive summary (EEA, 2024)

Area	Risks
Ecosystems	Coastal ecosystems Marine ecosystems Development of invasive species Changes in biodiversity, changes in species distributions Soil health Aquatic and wetland ecosystems Impact of forest encroachment (deforestation)
Food	Crop production and its products Fisheries and aquaculture Rising food supply prices Livestock and livestock products
Health	Heatwaves—for the general public health Fires in urbanised areas Threats to the well-being of the population due to unadjusted structures Disease agents in coastal waters Infectious diseases
Infrastructure	Thaw and river floods Coastal flooding Damage to infrastructure and buildings Power cuts due to heat and drought Power outages due to flooding Maritime and land transport
Economy/finance	European solidarity mechanisms Public finance Real estate and insurance markets Economics of water scarcity Medicines supply chains Raw material supply chains Financial markets Winter tourism

that global cereal yields have declined by an average of around 2–5% over the last four to five decades (Mirzabaev et al., 2023), with climate change expected to increase cereal prices by 1–29% in 2050 (Mbow et al., 2019). It is argued that, in the future, climate change will mean that some regions (tropics and subtropics) will suffer losses, while others (e.g. mid-latitudes) will be positively affected due to changes in climate conditions. Higher temperatures, especially in tropical and semi-tropical regions, are projected to reduce yields, reduce product quality, and increase food losses. This has immediate and long-term impacts on livelihoods of poor and vulnerable communities, contributing to greater risk of food insecurity which can become a driver of internal and external migration (Mbow et al., 2019).

Health. A particular health concern is the phenomenon of extreme weather—heat waves with negative consequences for society. In the summer of 2003, more than 70,000 excess deaths were recorded in 12 European countries. Long heat waves (>5 days) cause 1.5 to 5 times more negative impacts than shorter heat waves (less

than 5 days). Since the risk of mortality has been found to increase by between 0.2 and 5.5% with each additional degree (1 °C) above the region's temperature threshold, it is likely that rising ambient air temperatures will lead to a higher number of heat-related deaths (Radišauskas et al., 2014).

Schug et al. (2023) identify three main aspects of the health impacts of climate change. First, global warming has direct impacts on human health through increased average temperatures, changing precipitation patterns, droughts, floods, fires, and other more frequent and severe extreme weather events. All of these can lead to physical injury, death, and mental health problems. Secondly, health will be strongly influenced by changing local ecosystems, through changes in vector-borne diseases, air quality, and water pollution, and the prevalence and distribution of water- and food-borne diseases, which can directly affect health outcomes related to human nutrition or hygiene. This could change the known patterns of infectious disease transmission and contribute to the emergence of tropical and subtropical disease species in the region. Thirdly, health impacts will be strongly influenced by social factors, through impacts on different areas of the social system, such as increased poverty, migration, and displacement, changes in food production, water management, and potential food shortages. Together, these changes are contributing to the progressive vulnerability of global health.

Infrastructure. Threats to infrastructure (buildings, ports, roads, communications, energy supply, etc.) are mainly posed by rising sea levels, floods, high winds, high temperatures, forest fires, etc. Sea-level rise poses a major threat to coastal populations, as it will increase the frequency and intensity of flooding from storms, extreme floods, wave and backwater action, as well as intensify coastal and riverbank erosion, salinisation of water and soil, and, in some cases, permanent flooding. It is estimated that it will destroy entire population centres and force people to migrate, reduce agricultural production, threaten water infrastructure, roads, etc. (Duijndam et al., 2022; Motevalli et al., 2023). Coastal flooding in Europe is projected to affect 3.9 million people annually by 2100, with damages of around €1.6 trillion (Vousdoukas et al., 2020). In addition, strong winds and heavy rainfall caused by climate change, and landslides that form can devastate infrastructures that are technologically older and not adapted to the current effects of climate change. This can lead not only to temporary loss of housing, but also to disruptions in water and electricity supplies, loss of transport and communications, and large-scale population migrations.

Economics. Dellink et al. (2019) use economic forecasting models to estimate the economic consequences of climate change up to 2060 through specific drivers of economic growth, including labour productivity, capital stock, and land supply, as well as its indirect effects on other economies. The results showed that damage will increase twice as fast as global economic activity, with annual losses in global gross domestic product projected to be between 1.0% and 3.3% by 2060. The most significant negative economic effects will be on labour productivity and agriculture, with the increase in damages caused by sea-level rise being recorded after

mid-century. However, the damage to energy and tourism will not be significant on a global scale, as the benefits in some regions (countries in colder climates) will outweigh the damage in other regions. Extreme natural events such as high winds, heavy and sudden rainfall, floods and droughts will have an undeniable impact on the economy and social factors. In 1980–2013, economic losses due to extreme weather were estimated at around €400 billion in Europe (Suits et al., 2023).

9.4 Solutions for Building Climate Resilience

All of the risk areas analysed above are very closely interlinked and interdependent. Climate change and more intense and more frequent extreme weather events are changing growing conditions and crop yields, changing land use, threatening terrestrial and marine biodiversity, and altering animal migration patterns. Extreme weather events can disrupt supply chains and destroy infrastructure and, in extreme cases, lead to displacement or migration due to the critical impacts of climate change. It is not excluded that this puts stress on communities, which has a direct impact on people's psychological health. There are also direct economic costs, as the economic costs of dealing with the problems encountered (e.g. reconstruction of destroyed or flood-affected areas, damage to agricultural crops due to drought or other meteorological phenomena, etc.) can be very high. This requires countries to take measures to mitigate and adapt to the changes and impacts of climate change (Table 9.2).

A review of climate change vulnerabilities and management solutions suggests that climate change adaptation and resilience building must be approached in a systematic manner. The natural, social, and economic elements of the system are subject to the complex impacts of climate, and only a holistic and systemic approach by countries will allow for coherent policies to manage risks and make further decisions at the system-wide level.

9.5 Climate Resilience in the Baltic States

The Baltic States—Lithuania, Latvia, and Estonia—face similar risks and impacts from climate change as most European countries. However, given the specificities of small states (economic capacity, complexity of climate change outcomes, etc.), it is appropriate to analyse in more detail the impacts of climate change on the Baltic States and possible strategies for building resilience.

The drivers of climate change in the Baltic States are: sea-level rise, changes in temperature and precipitation, and the occurrence of extreme weather events (Climate…, 2017; Latvian…, 2019; Resolution…, 2021). The consequences are mostly in the same areas as those in the selected EU countries: natural ecosystems, food, health, infrastructure, and the economy.

Table 9.2 Solutions for building climate resilience by risk area in EU countries. (EEA, 2024; European Commission, 2021, 2024)

Risk area	Solutions for building climate resilience
Ecosystems	Comprehensively assessing natural capital; protecting forest imbalances; promoting forest fire risk prevention through technologically efficient monitoring systems; strengthening marine spatial planning and coastal management plans while preserving the diversity of marine ecosystems; fostering cooperation between countries on spatial planning and nature restoration plans; and strengthening the collective capacity of societies to more economically and efficiently manage and exploit water resources.
Food	Strengthening soil monitoring; conducting a study on agricultural adaptation to climate change; reassessing fishing quotas in line with realistic fishing opportunities; promoting sustainable agricultural models; assessing the possibility of drought-tolerant plant varieties, etc.
Health	Strengthening the European Climate and Health Observatory to help prepare local and national health systems for the effects of climate change; strengthening cross-border mobilisation of staff and relocation of patients; and developing new medicines and vaccines for emerging infectious diseases (e.g. tropical diseases).
Infrastructure	Reviewing and improving the resilience of infrastructure to climate change, as this determines the environmental resilience of both buildings and their occupants (sustainable building ventilation; healthier living environments and energy efficiency); increasing the scale of renovation and strengthening infrastructure standards; and reinforcing the readiness of the energy sector to cope with the risks of climate change.
Economy (finance)	Promoting insurance with resilience-building provisions that simultaneously promote vulnerability reduction, providing affordable insurance and reducing pressure on public finances after extreme events.

Natural ecosystems. The Baltic Sea coast is perhaps the most vulnerable region in the Baltic States to climate change. Recently, significant changes in water temperature, oxygen concentration, and acidity (Kilpys et al., 2017) have been identified as a key driver of changes in physical parameters. This has implications for water stratification and sea ice, which in turn affect the entire marine ecosystem. This could lead to an acceleration of the eutrophication process, changes in the dynamics of phytoplankton and zooplankton, threats to the diversity of fish species, seasonal fluctuations in water quality indicators, etc. In addition, the Curonian Lagoon will be affected by rising water levels, with parts of the shoreline flooded and the entire ecosystem of the lagoon changing as a result of more frequent inflows of saline water.

Natural ecosystems are adversely affected by strong winds and waves which erode the Baltic Sea shoreline, wash away sand, and destroy natural plant and bird breeding sites. For example, in the winter of 2022, four strong storms in quick succession washed away three times more sand from the beaches than Storm Erwin 17 years ago or Storm Felix in 2015. According to scientists, a total of 1.75 million cubic metres of sand were washed away, both in the Curonian Spit and on the mainland, which is almost a tenth of the total Curonian Spit dune mass (Šimanauskienė et al., 2022; Falaleeva et al., 2023).

Climate change effects such as reduced snow cover and changes in the amount and seasonality of precipitation are likely to reduce river flooding in spring, which in turn will reduce river runoff and nutrient loads (Povilaitis et al., 2018).

Food. Climate change has a direct impact on agriculture which is an important sector of the Baltic economy. Prolonged droughts, unpredictable rainfall, and other extreme weather events can lead to lower yields and higher food prices. Analysis of climate data in the Baltic States shows that average annual temperatures are rising steadily. By 2100, the average temperature in the Baltic region will rise by 1.5–5.5 °C (Avotniece et al., 2017; Kilpys et al., 2017). This will lead to milder winters, fewer days with snow cover, and more frequent thaws. The number of hot days above 30 °C in summer will increase significantly. Changes in the distribution of precipitation will also be recorded. Rainfall is forecast to increase by 3.7% and 13.5%. In Estonia, for example, a 30% increase in rainfall intensity has already been observed and is projected to increase further (Suits et al., 2023). Snow will be replaced by more frequent rain showers and more frequent downpours will be observed. This will lead to seasonal changes, with milder and shorter winter seasons, earlier spring onset, and longer and hotter summers (Rutgersson et al., 2022). The lengthening of the summer season will increase the length of the growing season which can favour agricultural development and the cultivation of new plants or crops. The effect of a longer growing season will depend on the amount of rainfall and temperature during the growing season. In Estonia and Latvia, yields are estimated to increase at the rate between 0 and + 30%, but in Lithuania, yields are projected to decrease between −5% and 0% (Falaleeva et al., 2023).

Health. According to Jaagus et al. (2024), the number of extremely warm days in the Baltic States approximately doubled between 1951 and 2021. This has increased the likelihood of heat waves which have a negative impact on human health. They are particularly relevant in cities, where "urban heat islands" occur, where the air in cities, and especially in the city centre, becomes hotter than in suburban or rural areas (Radišauskas et al., 2014). This can lead to the consequences discussed above, such as increased mortality, impaired mental well-being, or poor health, especially for vulnerable groups such as the elderly and those with pre-existing medical conditions (Falaleeva et al., 2023). Radišauskas et al. (2014) analysed 23 heatwave periods in Vilnius over the period from 1993 to 2007 and found that the mortality rates in Vilnius increase during severe heat waves. In Riga, heatwaves increased mortality from 12% to 22% in the ≥65 age group (Pfeifer et al., 2020). Heat waves can cause dehydration and heat stroke, which can be life threatening. It is important to ventilate buildings and reduce indoor temperatures during prolonged heat waves to reduce the risk of heat-related illness and death.

Heat waves have also been shown to accelerate the spread of various infectious diseases (Lian et al., 2023). In the Baltic States, as in Europe as a whole, rising temperatures and more frequent heat waves have changed the geographical distribution of vector-borne diseases, such as those spread by ticks and mosquitoes. These changes may lead to the emergence of diseases that were previously uncommon in this region, as warmer temperatures allow vectors to breed at higher latitudes (Nichols et al., 2014).

Increasingly frequent hurricane-force winds and tornadoes in the Baltic States also increase the likelihood of accidents, injuries, and trauma.

Infrastructure. Scientists (Kont et al., 2003; Bukantis et al., 2007; Kilpys et al., 2017, Meier et al., 2022) argue that the Baltic coastline, its infrastructure and the local population are mainly affected by sea-level rise, changes in water balance, increasing storms and hurricane winds, and changes in the ice regime. According to the maximum water level rise scenario, the average winter water level on the Lithuanian coast could rise by up to 100 cm during the twenty-first century (Bukantis et al., 2007). Cities and port areas located on the coast are projected to be at greatest risk of flooding. When analysing the coastlines of Lithuania, Latvia, and Estonia, it is argued that Estonia, which has the longest coastline (3794 km), will face the greatest challenges in terms of sea-level rise. In Estonia, the greatest economic risk from sea water rise is predicted to be in urban areas, especially in the Estonian capital Tallinn, where roads, houses, and other structures are often located very close to the current shoreline. The biggest ecological threat to the environment of the Gulf of Finland and the Baltic Sea as a whole is posed by the landfill site of the former uranium enrichment plant in Sillamäe, which is located very close to the shoreline and can be easily affected by storms (Kont et al., 2003). In Lithuania, the most significant rise will occur on the south-eastern coast of the Baltic Sea, where Klaipėda is located (predicted rise of 20–90 cm) (Bukantis et al., 2007; Kilpys et al., 2017).

Another pressing problem along the Baltic Sea coastline in the Baltic States is the erosion of shallow and sandy shores and the overtopping of sand caused not only by sea water rise, but also by strong winds and swell (Rutgersson et al., 2022). Recently, cyclonic activity has been intensifying, which leads to an increase in severe storms off the Baltic coasts in winter and spring, and shorter periods of calm weather in summer. Rising air temperatures, resulting in warmer winters, are also contributing to shoreline abrasion. This leads to the loss of coastal ice which is a natural breakwater that protects the coast from winter storms (Meier et al., 2022; Najafzadeha & Soomerea, 2024). This increases erosion and causes changes in the formation of sandy beaches and dunes. For the above reasons, it is predicted that the seafront will not be able to recover its equilibrium profile and may have negative impacts on biodiversity (threat of extinction of rare plant species, destruction of bird breeding sites, etc.) as well as on recreation and tourism (e.g. disappearance of beaches) in the long term (Soomere, 2024).

According to Nazarenko et al. (2023), studies in Lithuania do not show clear trends of increasing aridity and extreme droughts before 2000, but recent dry and warm summers have led to large changes in river discharge. For three consecutive summers between 2018 and 2020, hydrological droughts have been declared across the country occurring during the summer period. Total annual runoff may increase slightly due to higher rainfall, but this will not fully compensate for the loss and will have a negative impact on hydropower generation. On the other hand, it encourages more intensive use of solar and wind energy.

Strong hurricane-force winds can cause large fluctuations in sea level. For example, north-western Estonia has recorded the most extreme water level rise ever recorded (around 1.5 m) during a storm (Soomere et al., 2015), with significant

damage not only to the coastline but also to its infrastructure and ecosystems. A similar situation was recorded in January 2005, when Windstorm Erwin struck. The strong winds not only broke up forests and devastated the coastline, but also flooded the Old Town of Klaipėda and raised the water level in the Klaipėda channel by up to 1.5 metres (Ministry…, 2023). It took 6 days to restore power after the storm, which left around 150–180,000 people without electricity across the country.

Economy. Atmospheric changes increase the intensity of storms and strong winds which cause direct damage to buildings, coastlines, forests, etc. It is particularly damaging to forests. Dubovskis et al. (2019) point out that over the past decade, wind storms have caused direct economic losses of around €164 million to forest owners in Latvia alone. Extreme weather, temperature, and rainfall changes lead to lower agricultural and fisheries productivity, which reduces overall output and can have a negative impact on both the Baltic domestic market and exports.

In summary, all areas are interlinked, which only confirms the fact that climate resilience encompasses a combination of natural, social, and economic elements. For example, a reduction in summer rainfall increases the likelihood of droughts and thus the risk of fires; agriculture faces the potential loss or reduction of yields due to weather changes and extreme events, which threatens food security and economic losses; intense summer rains, heat and cold waves will force the review and adaptation of infrastructure in settlements (water collection systems, roads, ventilation/heating systems in buildings, etc.) as well as assessment of the potential risk of landslides (Kilpys et al., 2017). Therefore, monitoring and assessing climate change trends in a systemic approach is essential to enhance the resilience of the Baltic States to climate change.

The Notre Dame Global Adaptation Initiative conducts an annual study to compile an index of the countries most prepared to adapt to climate change (ND-GAIN Country Index). The assessment methodology is based on the assessment of the following two key parameters of climate change adaptation (ND-Gain, 2024):

- Country's vulnerability: It analyses a country's global impact, vulnerability, and adaptive capacity to the adverse effects of climate change in the following sectors: food, water, health, ecosystem service, human habitat, and infrastructure. A total of 36 indicators are assessed.
- Country's readiness: It assesses a country's economic, governance, and social readiness, i.e. its ability to leverage investments and turn them into adaptation actions. A total of 9 indicators are measured.

After looking at all the indicators, the researchers rank over 180 countries (including the Baltic States) in descending order. The countries with the highest scores and the best readiness to deal with the effects of climate change are ranked first.

Among the Baltic States, Estonia is the leading country with the highest climate change readiness index (Table 9.3). However, the differences between the Baltic States are not significant. Analysing the changes over time, it can be said that since 1995, when the index was launched, Lithuania has made the biggest leap in terms of

Table 9.3 Baltic Climate Change Readiness Index, distribution of vulnerability and readiness scores (based on the data of Notre Dame Global Adaptation Initiative (ND-Gain, 2024a))

Index in ND-GAIN Country Index			Vulnerability score 0–1 (the lower the rating, the better)			Readiness score 0–1 (the higher the rating, the better)		
Ranked among 180 countries	Country	Score	Location	Country	Score	Location	Country	Score
21	Estonia	64.5	27	Estonia	0.330	24	Estonia	0.621
28	Lithuania	61.7	47	Lithuania	0.361	27	Lithuania	0.594
31	Latvia	60.8	50	Latvia	0.364	31	Latvia	0.581

improvement, rising from the 42nd to the 28th place, followed by Estonia from the 30th to the 21st, and Latvia from the 37th to the 31st compared to the rest of the world. These trends show that the Baltic States are taking a responsible approach to building climate resilience. However, given the imminent impacts of climate change in the short term, it is necessary to further plan and use management strategies to help mitigate and adapt to the effects of climate change.

The climate change management policies of the Baltic States are formulated and implemented in accordance with international agreements such as the United Nations Framework Convention on Climate Change, the Paris Agreements with commitments for the period from 2021 to 2030, and the United Nations 2030 Agenda for Sustainable Development, etc. Governance decisions are therefore broadly in line with those implemented by the European Union (Table 9.2).

When analysing the Baltic adaptation plans (Climate…, 2017; Latvian…, 2019; Resolution…, 2021), it is useful to analyse resilience to climate change through the concepts of absorptive, adaptive, and transformative capacities, as identified by Isaaka et al. (2023).

The need to increase the absorptive capacity in the Baltic States is directly linked to extreme climate events. At the heart of this resilience building are quick and short-term solutions to reduce damage in the event of disasters. Prevention therefore focuses on the direct impact on societies and economies through technology, infrastructure reinforcement, and the provision of crisis management measures. In order to achieve higher resilience, the Baltic States are upgrading air quality and early warning station networks and laboratories; adapting urban sewerage systems to cope with heavy rainfall and flash floods; modernising electricity grids and improving electricity redistribution systems to avoid power outages; strengthening civil defence and emergency management capacities to respond effectively to floods, forest fires, heat waves, etc.

Adaptive capacity refers to adapting to ever-changing climatic conditions and coping with their effects. In the Baltic States, this includes changes in the agricultural sector, upgrading urban and coastal infrastructure, modernising water management systems, etc. In this context, the Baltic States are investing in the conservation and restoration of wetlands, peatlands, and forests (e.g. LIFE ForestHabitatLT project); investing in the adaptation of infrastructure and the development of green urban spaces to help buffer against extreme weather events such as high winds or

heat waves; and investing in the protection of the Baltic Sea coastline (e.g. Baltic Sea2Land project). The Baltic governments are financing the renovation of apartment buildings for energy consumption and efficiency; promoting the use of renewable energy sources (solar, geothermal, biofuels) through climate change programmes; and strengthening the management structure through governance policy strategies, implementation plans and programmes to ensure synergies and cooperation not only between the responsible authorities, but also between countries.

Transformative capacity of climate resilience in the Baltic States means introducing lasting and fundamental changes in economic, energy, and social systems in order not only to adapt to current circumstances as part of building adaptive capacities, but also to introduce new strategies and tools that will be effective in a changing climate. The Baltic States are investing in renewable energy (wind, solar, and biomass); raising public awareness and education through events, specialised publications, etc.; participating in international projects (Interreg Baltic Sea Regions, CEforestry, Baltic Biodiversity, etc.); and developing new technologies and tools to address climate change. The focus is on improving the Baltic Sea ecosystem (preserving biodiversity, reducing pollution, regulating sea levels) and natural resource management through international cooperation (HELCOM, Baltic Sea Regions (BHR), Protect Baltic, etc.).

In summary, building resilience to climate change in the Baltic States is a complex process that requires an integrated approach across ecological, social, and economic systems. To be effective in this context, the concept of resilience needs to be understood holistically, taking into account the absorptive, adaptive, and transformative capacities of countries. The Baltic adaptation plans focus on measures aimed at building adaptive capacity, such as increasing the resilience of the agriculture, forestry, fisheries, and tourism sectors; flood risk management and the development of green spaces and urban landscapes and infrastructure, etc. They also highlight the need to develop policy instruments and financial instruments to support resilience-building initiatives in order to ensure sustainable development and climate resilience in the Baltic States.

9.6 Conclusions

Climate resilience is a broad concept encompassing not only the ability of countries and societies to respond and adapt to changing climatic conditions and their effects, but also to provide mitigation measures. Effective resilience building requires strengthening the absorptive, adaptive, and transformative capacities of societies, which is achieved through the sustained and systematic implementation of measures in the social, economic, and political spheres.

The areas most affected by climate change in the Baltic States are the natural ecosystems, agriculture, infrastructure, health and economic sectors. Therefore, the systematic assessment of impacts, identification of risks, and design of risk reduction and adaptation measures help increase the resilience of societies and countries to the potential threats posed by climate change. In this context, states should look

at resilience in a broader sense and assess the absorptive, adaptive, and transformative capacities of resilience in equal measure. The Baltic Climate Resilience Analysis confirmed that all three groups are closely interlinked, but the intended outcome and measures differ.

Given the increasing frequency of extreme weather events in the Baltic States, it is appropriate to focus more on developing absorptive resilience capacities, i.e. the means to respond quickly and effectively to the effects of such events. In the Baltic States, these include strengthening infrastructure and adapting to climate change, setting up monitoring and warning systems, improving crisis management systems and tools, public information, etc.

The Baltic States are focusing on the development of adaptive capacities which includes potential strategies and measures to adapt to ever-changing climate conditions and their effects: restoring natural ecosystems, developing green urban areas, investing in research, strengthening governance structures for managing climate crises, developing a financial support system, etc.

Transformative capacity involves long-term changes in a country related to climate change, and is usually realised through the implementation of new strategies. In the Baltic States, this is to be achieved in the context of climate change through investing in renewable energy, raising public awareness and education, promoting regional cooperation between the Baltic States and the EU on climate action, knowledge sharing and joint infrastructure projects, attracting private investment, etc. In the long term, the properly selected and implemented transformative resilience measures are likely to increase both adaptive and absorptive capacities, which in turn will pave the way for sustainable development and climate resilience in the Baltic States.

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